# Exam in Signals & Transforms 2025-08-19

1. a)

$$z(t) = x(t+0{,}5) * y(t)$$

$$= \sin(\pi(t+0{,}5)) * \mathrm{rect}\left(\tfrac{t}{2}\right)$$

$$= \sin\left(\pi t + \tfrac{\pi}{2}\right) * \mathrm{rect}\left(\tfrac{t}{2}\right)$$

$$= \cos(\pi t) * \mathrm{rect}\left(\tfrac{t}{2}\right)$$

$$\Updownarrow$$

$$Z(\omega) = \pi\left[\delta(\omega-\pi) + \delta(\omega+\pi)\right] \cdot 2\,\mathrm{sinc}\left(\frac{2\omega}{2\pi}\right)$$

$$= 2\pi\left[\delta(\omega-\pi) + \delta(\omega+\pi)\right]\mathrm{sinc}\left(\frac{\omega}{\pi}\right)$$

$$= 2\pi\left[\delta(\omega-\pi)\,\mathrm{sinc}\left(\frac{\omega}{\pi}\right) + \delta(\omega+\pi)\,\mathrm{sinc}\left(\frac{\omega}{\pi}\right)\right]$$

$$= 2\pi\left[\delta(\omega-\pi)\,\mathrm{sinc}\left(\frac{\pi}{\pi}\right) + \delta(\omega+\pi)\,\mathrm{sinc}\left(\frac{-\pi}{\pi}\right)\right]$$

$\delta(\omega-\omega_0) \neq 0$ only for $\omega = \omega_0$

$$= 2\pi\left[\delta(\omega-\pi)\,\mathrm{sinc}(1) + \delta(\omega+\pi)\,\mathrm{sinc}(-1)\right]$$

$\mathrm{sinc}(1) = \mathrm{sinc}(-1)$ (even)

$$= 2\pi\,\mathrm{sinc}(1)\left[\delta(\omega-\pi) + \delta(\omega+\pi)\right]$$

$$= 2\pi \cdot 0 \cdot \left[\delta(\omega-\pi) + \delta(\omega+\pi)\right]$$

$\mathrm{sinc}(x) = \dfrac{\sin(\pi x)}{\pi x}$

$$\underline{\underline{Z(\omega) = 0}}$$

(This is due to the fact that $y(t)$ has the same width as one period of $x(t)$, which yields that the convolution of the two is an integral over one full period of $x(t)$.)

b) For example:

- The odd component's Fourier transform is purely imaginary and thus only consists of sine components.
- Vice-versa, the even component's Fourier transform is purely real. Thus, it only consists of cosine components.

c) Unfortunately, we do not know the Fourier transform of the unit step (Heavyside function). However, $v(t)$ is causal and thus, we can get the Fourier transform from the Laplace transform.

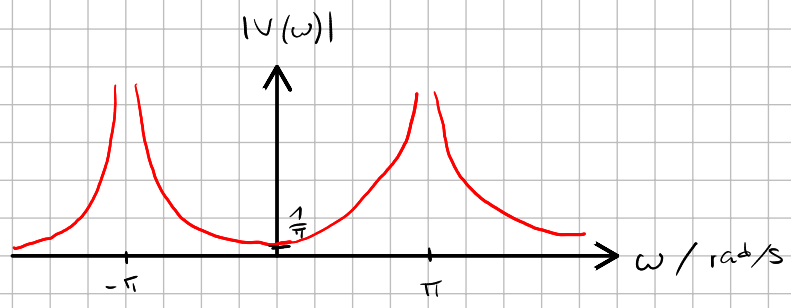
$$v(t) = x(t)u(t) = \sin(\pi t)u(t)$$

$\Updownarrow$ Causal sine with $\omega_0 = \pi$ rad/s

$$V(s) = \frac{\pi}{s^2 + \pi^2}$$

$$V(j\omega) = V(s)\big|_{s=j\omega} = \frac{\pi}{(j\omega)^2 + \pi^2} = \frac{\pi}{\pi^2 - \omega^2}$$

$\Rightarrow$ This corresponds to a 2nd order system with poles in $\pm j\pi$. Thus, we have no damping and resonance frequency $\omega_0 = \pi$ rad/s.

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2a) Rewrite $H(s)$ to Bode form:

$$H(s) = 5\frac{s+1}{(s+0{,}1)(s+10)} = \frac{5}{0{,}1\cdot 10}\frac{s+1}{(s/0{,}1+1)(s/10+1)}$$

$$= 5\frac{s+1}{(s/0{,}1+1)(s/10+1)}$$

Hence, we have the following four terms:

1. A constant $K = 5$:

$\Rightarrow 20\log_{10}(5) = 14$ dB

2. A first order term with breakpoint in $\omega = 1$ rad/s in the numerator, $s+1$:

$$20\log_{10}(|j\omega+1|) = \begin{cases} 0 \text{ dB} & \omega \ll 1 \text{ rad/s} \\ +3 \text{ dB} & \omega = 1 \text{ rad/s} \\ +20 \text{ dB/dec.} & \omega \gg 1 \text{ rad/s} \end{cases}$$

$$\angle(j\omega+1) = \begin{cases} 0^\circ & \omega \ll 1 \text{ rad/s} \\ +45^\circ & \omega = 1 \text{ rad/s} \\ +90^\circ & \omega \gg 1 \text{ rad/s} \end{cases}$$

3. A first order term with breakpoint 0,1 rad/s in the denominator, $\frac{1}{s/0{,}1+1}$:

$$20\log_{10}\left(\frac{1}{|j\omega/0{,}1+1|}\right) = \begin{cases} 0 \text{ dB} & \omega \ll 0{,}1 \text{ rad/s} \\ -3 \text{ dB} & \omega = 0{,}1 \text{ rad/s} \\ -20 \text{ dB/dec} & \omega \gg 0{,}1 \text{ rad/s} \end{cases}$$

$$\angle \frac{1}{j\omega/0{,}1+1} = \begin{cases} 0^\circ & \omega \ll 0{,}1 \text{ rad/s} \\ -45^\circ & \omega = 0{,}1 \text{ rad/s} \\ -90^\circ & \omega \gg 0{,}1 \text{ rad/s} \end{cases}$$

4. A first order term with breakpoint 10 in the denominator, $\frac{1}{s/10+1}$:

$$20\log_{10}\left(\frac{1}{|j\omega/10+1|}\right) = \begin{cases} 0 \text{ dB} & \omega \ll 10 \text{ rad/s} \\ -3 \text{ dB} & \omega = 10 \text{ rad/s} \\ -20 \text{ dB/dec} & \omega \gg 10 \text{ rad/s} \end{cases}$$

$$\angle \frac{1}{j\omega/10+1} = \begin{cases} 0^\circ & \omega \ll 10 \text{ rad/s} \\ -45^\circ & \omega = 10 \text{ rad/s} \\ -90^\circ & \omega \gg 10 \text{ rad/s} \end{cases}$$

This yields the Bode diagram at the end.

b) We can check the poles of the transfer function to determine BIBO-stability:

$$H(s) = 5\,\frac{s+1}{(s+0{,}1)(s+10)}$$

Poles (roots of the denominator):

$(s+0{,}1)(s+10) = 0$

1. $s = -0{,}1$
2. $s = -10$

The system is stable if all poles satisfy $\mathrm{Re}\{p_i\} < 0$.

We have that $\underline{\underline{p_1 = -0{,}1,\ p_2 = -10 \Rightarrow \text{Stable}}}$

c) We can use the laplace-domain input-output relationship:

$$Y(s) = H(s)\,X(s)$$

where

$$x(t) = u(t) \Leftrightarrow X(s) = \frac{1}{s}$$

Thus,

$$Y(s) = 5\,\frac{s+1}{(s+0{,}1)(s+10)} \cdot \frac{1}{s} = 5\,\frac{s+1}{s\,(s+0{,}1)\,(s+10)}$$

Partial fraction expansion:

$$Y(s) = 5\left(\frac{a}{s} + \frac{b}{s+0{,}1} + \frac{c}{s+10}\right)$$

$$= 5\,\frac{a(s+0{,}1)(s+10) + sb\,(s+10) + cs\,(s+0{,}1)}{s\,(s+0{,}1)(s+10)}$$

Considering the numerator only:

$$a(s+0{,}1)(s+10) + sb(s+10) + cs(s+0{,}1)$$

$$= a(s^2 + 10{,}1s + 1) + s^2 b + 10sb + s^2 c + 0{,}1sc$$

$$= as^2 + 10{,}1as + a + s^2 b + 10sb + s^2 c + 0{,}1cs$$

$$= s^2(a+b+c) + s(10{,}1a + 10b + 0{,}1c) + a$$

Comparison of the coefficients:

① $s^0$: $\underline{a = 1}$

② $s^1$: $10{,}1a + 10b + 0{,}1c = 1$

$$10{,}1 + 10b + 0{,}1c = 1$$

$$10b + 0{,}1c = -9{,}1$$

$$b = -\frac{1}{10}(9{,}1 + 0{,}1c)$$

$$= -0{,}91 - 0{,}01c$$

③ $s^2$: $a + b + c = 0$

$$1 - 0{,}91 - 0{,}01c + c = 0$$

$$0{,}09 + 0{,}99c = 0$$

$$c = -\frac{100}{99} \cdot \frac{9}{100} =$$

$$\underline{c = -\frac{1}{11} \approx -0{,}0909}$$

Combining ② & ③:

$$b = -0{,}91 - 0{,}01c$$
$$= -0{,}91 - 0{,}01 \cdot (-0{,}091)$$
$$= -0{,}91 + 0{,}0009$$
$$\underline{\approx -0{,}91}$$

$$\Rightarrow \underline{a = 1,\ b \approx -0{,}91,\ c \approx -0{,}091}$$

Thus,

$$Y(s) = 5\left(\frac{1}{s} - \frac{0{,}91}{s+0{,}1} - \frac{0{,}091}{s+10}\right)$$
$$\Updownarrow$$
$$y(t) = 5\left(u(t) - 0{,}91\, e^{-0{,}1t} u(t) - 0{,}091\, e^{-10t} u(t)\right)$$
$$\underline{\underline{= 5\left(1 - 0{,}91\, e^{-0{,}1t} - 0{,}091\, e^{-10t}\right) u(t)}}$$

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3a) The order of a Butterworth filter is given by

$$N \geq \frac{\log\left(\sqrt{\frac{A_s^2 - 1}{A_p^2 - 1}}\right)}{\log\left(\frac{\omega_s}{\omega_p}\right)} = \frac{\log\left(\sqrt{\frac{(10^{45/20})^2 - 1}{(10^{2/20})^2 - 1}}\right)}{\log\left(\frac{6000}{2000}\right)} = 4{,}96$$

$$\Rightarrow \underline{\underline{N \geq 5}}$$

b) There are several challenges associated with that. For example, it makes it difficult to implement high-order filters as it would require large circuits. Furthermore, components come in predefined values and have tolerances, which leads to deviations from the original specifications and may even cause instability.

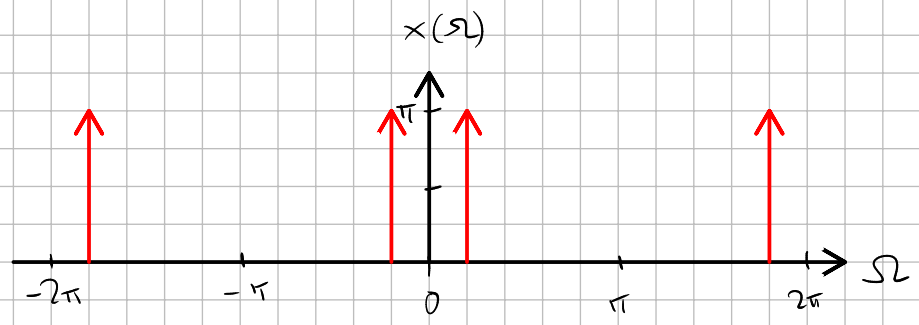
4. a) The signal is sampled with $T_s = 2s$, hence,

$$x[k] = x(kT_s) = \cos\left(\frac{2\pi}{20} kT_s\right) = \cos\left(\frac{2\pi}{20} k\, 2\right)$$

$$= \cos\left(\frac{2\pi}{10} k\right)$$

$$\Updownarrow$$

$$\underline{\underline{X(\Omega) = \pi\left[\delta\left(\Omega + \frac{2\pi}{10}\right) + \delta\left(\Omega - \frac{2\pi}{10}\right)\right]}} \quad \left(\Omega \in (-\pi, \pi);\ X(\Omega + 2\pi) = X(\Omega)\right)$$




b) The period of the sampled signal is $k_0 = 10$ samples. Hence, with $K = 20$ samples, we sample 2 periods perfectly. Furthermore, the spectral resolution is

$$\Delta\Omega = \frac{2\pi}{K} = \frac{2\pi}{20}.$$

Thus, $\Omega_0 = \frac{2\pi}{10}$ ends up on bin $l = 2$ and its copy at $2\pi - \frac{2\pi}{10}$ on bin $l = 18$.

Finally, for periodic signals we have that

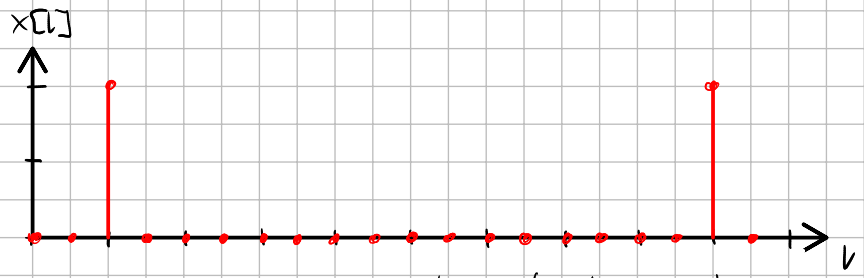
$$c_l \approx \frac{1}{K} X[l] \quad \Rightarrow \quad X[l] \approx K c_l$$

$$X(\Omega) = 2\pi \sum_{n=-\infty}^{\infty} c_n \delta(\Omega - n\Omega_0) \quad \Rightarrow \quad c_l = \frac{X(\Omega_l)}{2\pi}$$

which yields

$$X[l] = K \frac{X(\Omega_l)}{2\pi} = K \frac{\pi}{2\pi} = 10$$

This yields the DFT on the next page.




c) The common period of the two components is $T_0 = 60$ s. Since $t = 60$ s marks the start of the second period, we need to sample

$$\underline{\underline{T = T_0 - T_s = 58 \text{ s}}}$$

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5a) The impulse response is given by

$$h[k] \Leftrightarrow H(z)$$

with

$$H(z) = \frac{2 - 1{,}5 z^{-1}}{1 - 1{,}5 z^{-1} + 0{,}5 z^{-2}}$$

Note that this can not be rewritten into the form of a causal cosine as the denominator can not have coefficients 1 for $z^0$ and $z^{-2}$ at the same time. Hence, we need partial fraction expansion.

$$H(z) = \frac{2z^2 - 1{,}5z}{z^2 - 1{,}5z + 0{,}5} = \frac{2z^2 - 1{,}5z}{(z-1)(z-0{,}5)}$$

$$= \frac{az+b}{z-1} + \frac{cz+d}{z-0{,}5}$$

$$= \frac{(az+b)(z-0{,}5) + (cz+d)(z-1)}{(z-1)(z-0{,}5)}$$

Numerator only:

$$2z^2 - 1{,}5z = az^2 - 0{,}5az + bz - 0{,}5b + cz^2 - cz + dz - d$$

$$= z^2(a+c) + z(b - 0{,}5a - c + d) - 0{,}5b - d$$

① $z^2$: $\quad a + c = 2$

② $z^1$: $\quad b - 0{,}5a - c + d = -1{,}5$

③ $z^0$: $\quad -0{,}5b - d = 0$

$\Rightarrow$ We have 4 unknowns but only 3 equations. We will need to make a choice.

- From ①: $a = 2 - c$
- From ③: $d = -0{,}5b$
- From ②: $b - 0{,}5 \cdot (2-c) - c - 0{,}5b = -1{,}5$

$$0{,}5b - 1 + 0{,}5c - c = -1{,}5$$
$$0{,}5b - 0{,}5c = -0{,}5$$
$$b - c = -1$$
$$b = c - 1$$

$\Rightarrow$ Choose $\underline{c = 1 \Rightarrow b = 0,\ a = 1,\ d = 0}$

$$\Rightarrow H(z) = \frac{z}{z-1} + \frac{z}{z-0{,}5} = \frac{1}{1-z^{-1}} + \frac{1}{1-0{,}5z^{-1}}$$

$$\Updownarrow$$

$$h[k] = u[k] + 0{,}5^k u[k]$$
$$\underline{\underline{h[k] = (1 + 0{,}5^k)u[k]}}$$

b) From a), we have that the poles are $p_1 = 1$ and $p_2 = 0{,}5$. For a discrete-time LTI system, it is required that $|p_i| < 1$ for all poles. Since $|p_1| = 1$, the system is <u>not stable!</u>

c) The input can be written as

$$x[k] = \sum_{n=0}^{4} \delta[k-n]$$

Hence,

$$y[k] = x[k] * h[k]$$
$$= \sum_{n=0}^{4} \delta[k-n] * h[k]$$
$$= \sum_{n=0}^{4} h[k-n]$$

$$\underline{\underline{y[k] = \sum_{n=0}^{4} (1 - 0{,}5^{(k-n)}) u[k-n]}}$$

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6.a) The signal consists of two harmonic oscillations with frequencies

$$\omega_1 = \frac{2\pi}{0{,}1} \text{ rad/s} = 10 \cdot 2\pi \text{ rad/s}$$

$$\omega_2 = \frac{2\pi}{0{,}01} \text{ rad/s} = 100 \cdot 2\pi \text{ rad/s}$$

According to the Shannon-Nyquist sampling theorem, we thus have that

$$\omega_s \geq 2\omega_2 = 200 \cdot 2\pi \text{ rad/s}$$

$$\Rightarrow \underline{\underline{\omega_s \geq 400\pi \text{ rad/s}}}$$

b) With $\omega_s = 1000\pi$ rad/s, we have

$$\omega_u = \frac{\omega_s}{2} = 500\pi \text{ rad/s}$$

This should be the stopband edge frequency. Since the highest frequency is $\omega_2 = 200\pi$ rad/s, this can be chosen as the passband edge frequency.

A stopband attenuation of 40 dB reduces signals in the stobband by a factor of 100, whereas a passband ripple of 3 dB retains at least 50% of the energy in the passband.

$$\Rightarrow \begin{cases} \text{Passband edge frequency:} & 200\pi \text{ rad/s} \\ \text{Stopband edge frequency:} & 500\pi \text{ rad/s} \\ \text{Passband ripple} & 3 \text{ dB} \\ \text{Stopband attenuation:} & 40 \text{ dB} \end{cases}$$